

CSCI 545 Lab 3 - Report

Jailbreakers

Exercise Questions

Question 5

Reducing the goal precision from 1.0 to 0.2 significantly increased the difficulty of reaching the goal because the RRT must now enter a much smaller region of joint space before terminating. When using only uniform random sampling, the planner either took a very long time or failed to find a solution, since the probability of landing close enough to the goal purely by random exploration is extremely low. After adding goal-biased sampling the planner became efficient again even with the stricter tolerance. The RRT tree showed noticeably denser branching around the goal region, which helped guide extensions toward the target configuration. The resulting smoothed trajectory had a more accurate final pose and a cleaner approach to the goal compared to Question 4, although the overall path shape remained similar. The higher precision also makes the robot end in a configuration that avoids potential near-goal collisions.

Question 6

Calling the function with probability 1.0 is not a good idea because it eliminates the exploration phase of the RRT algorithm, causing the tree to only grow in the immediate proximity of the goal configuration. This approach fails when there are obstacles between the start and goal positions, as the algorithm would never explore the broader configuration space needed to find a path around these obstacles. For example, if there is an obstacle interpolated between the start and goal configurations, sampling exclusively near the goal would prevent the tree from ever exploring the intermediate space required to navigate around the obstacle, and the algorithm would fail to find a solution entirely. This problem would not occur with proper exploration-based RRT, which would naturally discover paths around such obstacles. The RRT algorithm relies on random exploration of the configuration space to discover feasible paths, and removing this exploration by only sampling near the goal defeats the fundamental purpose of the algorithm.

Team Contributions

The four members of our team made equal contributions to this lab (25% each). While each member began by independently setting up their environment and working through the exercises, the final report was a collaborative effort. We collated our individual results, discussed our findings, and jointly wrote the final document.

Resources Consulted

- Official documentation of ROS for more information on packages.
- ChatGPT for in-depth explanations and for some formatting help with LaTeX and debugging issues with the Docker images.